

Bluestock Mutual Fund Analytics Capstone Project

Final Report

Prepared By:

Aman Kumar Rai

1. Executive Summary

The Bluestock Mutual Fund Analytics Capstone Project was undertaken to build an end-to-end analytics solution for evaluating mutual fund performance, investor behaviour, and portfolio risk. The project combines multiple raw datasets covering fund master information, NAV history, AUM trends, SIP and category-wise inflows, portfolio holdings, investor transactions, and benchmark indices into a unified analytical framework using Python and Power BI.

The primary objective was to design a complete pipeline that extracts raw CSV data, transforms it into clean and structured datasets, and loads it into both Python (for advanced statistical and risk analysis) and Power BI (for interactive visual dashboards). This enables stakeholders to track fund performance against benchmarks, understand investor inflow patterns, and identify risk-adjusted return opportunities across the fund universe.

Beyond standard performance reporting, the project incorporates advanced analytics such as Value at Risk (VaR), Conditional Value at Risk (CVaR), rolling Sharpe ratio analysis, cohort-based investor analysis, investor risk segmentation, and a fund recommendation engine. The result is a comprehensive analytics suite that supports data-driven decision-making for both fund managers and investors.

2. Problem Statement

Mutual fund investors and analysts often struggle to evaluate fund performance holistically, as raw NAV and AUM data alone do not reveal risk-adjusted returns, investor behaviour patterns, or comparative performance against benchmarks. The core problem addressed by this project is threefold:

- **Mutual Fund Performance Analysis** - Evaluating returns (CAGR), risk-adjusted returns (Sharpe ratio), and downside risk (Maximum Drawdown) across a universe of mutual fund schemes.
- **Investor Behaviour Analysis** - Understanding SIP inflow trends, category-wise investment preferences, and investor transaction patterns over time.
- **Risk Analysis** - Quantifying portfolio risk using statistical measures such as VaR, CVaR, and rolling Sharpe ratios, and segmenting investors based on their risk profiles.

The goal is to consolidate these dimensions into a single analytics framework that is both statistically rigorous (via Python) and accessible to business stakeholders (via Power BI dashboards).

3. Data Sources

The project is built on the following raw datasets, each contributing a distinct dimension to the overall analysis:

Dataset	Description
Fund Master	Master list of mutual fund schemes with identifying details such as AMFI code, fund name, category, and tenure.
NAV History	Historical Net Asset Value (NAV) records for each fund, used to compute returns and risk metrics.
AUM Data	Assets Under Management figures tracked over time for each scheme, indicating fund size and growth.
SIP Inflows	Systematic Investment Plan inflow data, capturing recurring investor contributions over time.
Category Inflows	Aggregated inflow/outflow data segmented by mutual fund category (e.g., Equity, Debt, Hybrid).
Portfolio Holdings	Underlying securities and asset allocation held within each fund's portfolio.
Investor Transactions	Individual investor-level buy/sell/redeem transaction records.
Benchmark Indices	Index-level data (e.g., Nifty 50, Sensex) used as a reference for performance comparison.

4. ETL Process

A structured Extract-Transform-Load (ETL) pipeline was designed to convert raw, fragmented CSV files into analysis-ready datasets for both Python-based statistical modelling and Power BI dashboarding.

Extract: Raw CSV files corresponding to Fund Master, NAV History, AUM Data, SIP Inflows, Category Inflows, Portfolio Holdings, Investor Transactions, and Benchmark Indices were extracted from the source data folder. Each file was validated for schema consistency, encoding, and date formats before further processing.

Transform: The extracted datasets were cleaned and transformed - handling missing values, standardising date and numeric formats, merging fund identifiers (AMFI codes) across datasets, and engineering derived fields such as daily/annualised returns, rolling returns, CAGR, Sharpe ratio, and maximum drawdown. Investor transaction and SIP data were aggregated to derive cohort and inflow trend tables.

Load: The cleaned and transformed datasets were loaded into two destinations - Python (Pandas DataFrames) for advanced statistical computations such as VaR, CVaR, rolling Sharpe ratios, and the recommendation engine; and Power BI, where the data model was used to build the interactive dashboards covering fund performance, NAV trends, AUM growth, and investor analytics.

5. EDA Findings

Exploratory Data Analysis was performed to understand the distribution of fund returns, AUM trends, and investor inflow patterns. The interactive Power BI dashboard below summarises the key findings across the fund universe.

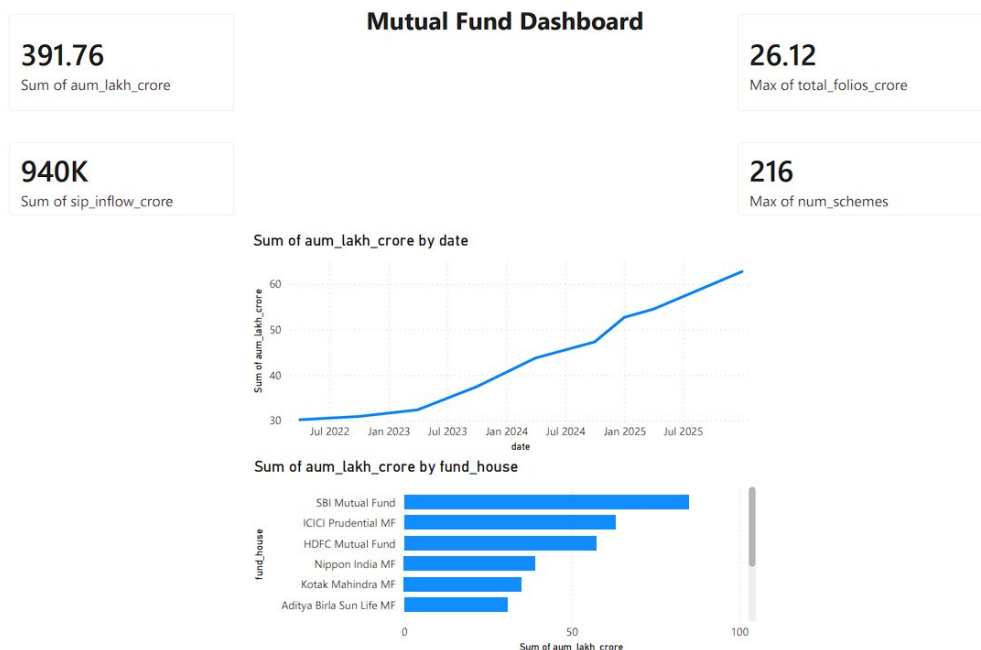


Figure 5.1: Bluestock Mutual Fund Analytics - Power BI Dashboard Overview

Across the 40 funds analysed, the average CAGR was approximately 18.6%, with a wide spread ranging from 1.3% to 36.6%, indicating significant variation in fund performance. The average

Sharpe ratio stood at 0.54, with 31 out of 40 funds (about 78%) showing a positive risk-adjusted return, while the remaining funds underperformed on a risk-adjusted basis.

6. Performance Analysis

NAV Performance: NAV history for each fund was analysed to compute compound annual growth rate (CAGR) and visualise growth trajectories over time. The chart below shows fund-level NAV performance trends.

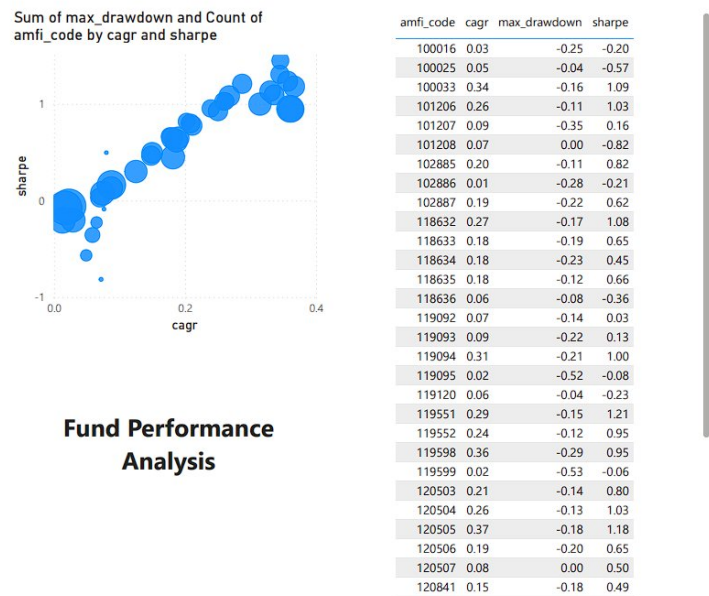


Figure 6.1: Fund-wise NAV Performance

Benchmark Comparison: Each fund's NAV-based returns were compared against relevant benchmark indices (e.g., Nifty 50) to assess whether the fund outperformed or underperformed the broader market over the analysis period.

NAV vs Benchmark Analysis

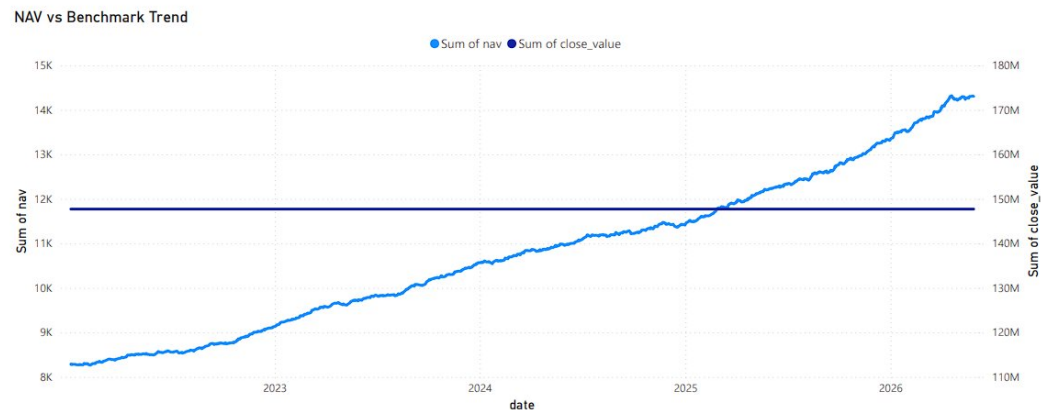


Figure 6.2: NAV vs Benchmark Comparison

Figure 6.3: Benchmark Comparison Summary

Investor Analytics: SIP inflows, category-wise inflows, and investor transaction data were analysed to identify investment trends, popular fund categories, and investor activity patterns over time.

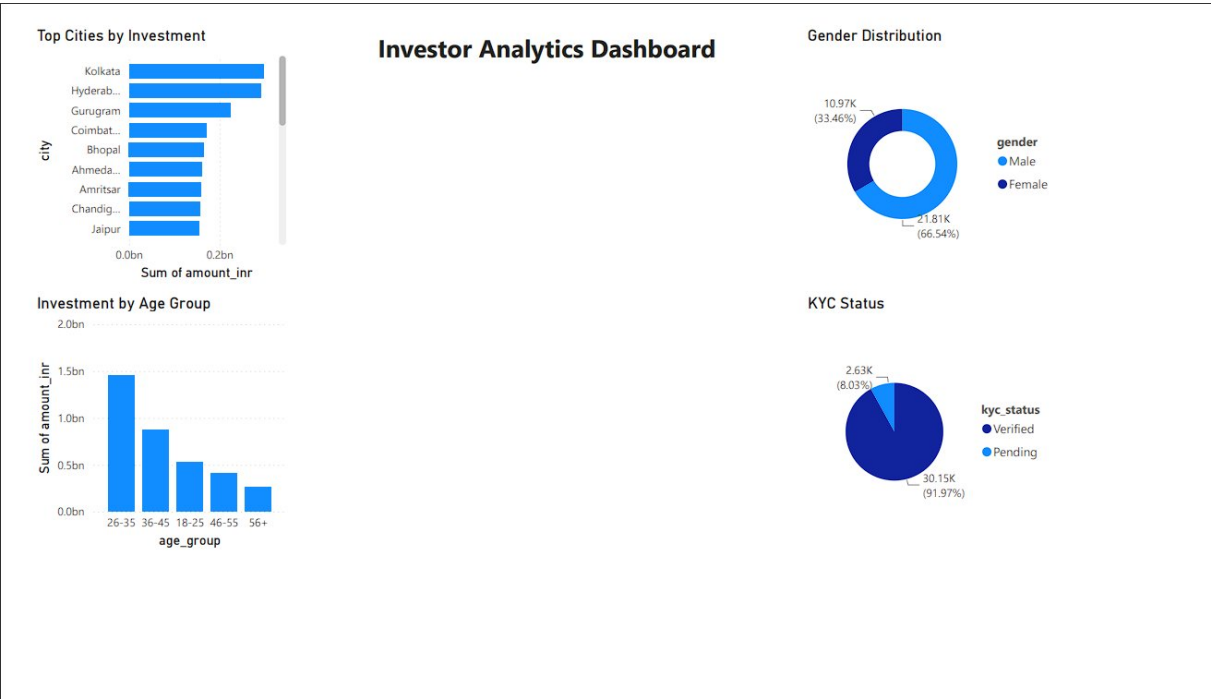


Figure 6.4: Investor Analytics Dashboard

7. Advanced Analytics

Value at Risk (VaR) and Conditional Value at Risk (CVaR): Historical NAV returns for each fund were used to estimate VaR (the maximum expected loss at a given confidence level) and CVaR (the expected loss beyond the VaR threshold). These metrics provide a quantitative view of downside risk exposure for each scheme, complementing traditional metrics like maximum drawdown, which ranged from -0.1% to as low as -52.6% across the fund universe.

Rolling Sharpe Ratio: A rolling-window Sharpe ratio was computed for each fund to track how risk-adjusted performance evolves over time, helping identify periods of consistent outperformance versus periods of elevated risk relative to returns.

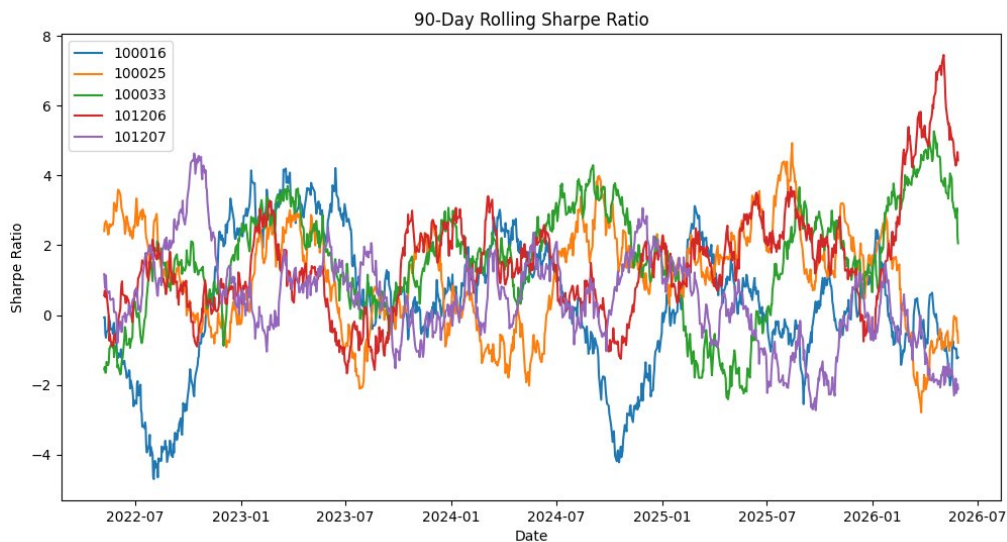


Figure 7.1: Rolling Sharpe Ratio Trend

Cohort Analysis: Investor transaction data was grouped into cohorts based on the period of first investment, enabling analysis of retention, contribution growth, and behaviour patterns across different investor cohorts over time.

Investor Risk Segmentation: Investors were segmented into risk profiles (e.g., Conservative, Moderate, Aggressive) based on their portfolio allocation across fund categories, transaction frequency, and historical exposure to high-volatility funds.

Recommendation Engine: A rule-based and similarity-driven recommendation engine was developed to suggest suitable mutual fund schemes to investors based on their risk segment, past investment behaviour, and the risk-return profile of available funds. Top-performing funds by Sharpe ratio (e.g., AMFI codes 148567, 120843, and 148569, each with a Sharpe ratio above 1.3) were prioritised for recommendation to investors with a higher risk appetite.

8. Limitations

- **Historical data only** - All analyses, including NAV performance, risk metrics, and recommendations, are based on historical data. Past performance trends may not be indicative of future fund behaviour.
- **No live market feed** - The current pipeline does not integrate with a live market data feed, meaning NAV values, AUM figures, and benchmark comparisons are not updated in real time and require periodic manual refreshes.

9. Recommendations

- **Add real-time NAV integration** - Connect the pipeline to a live NAV/market data API so that fund performance metrics, benchmark comparisons, and dashboards reflect up-to-date information.
- **Add ML-based recommendation system** - Enhance the current rule-based recommendation engine with a machine learning model trained on historical investor behaviour and fund performance to deliver more personalised and adaptive fund recommendations.

10. Conclusion

The Bluestock Mutual Fund Analytics Capstone Project has been completed successfully. A complete ETL pipeline was built to consolidate fund master, NAV, AUM, SIP, category inflow, portfolio holdings, investor transaction, and benchmark data into Python and Power BI. Performance analysis, benchmark comparison, and investor analytics dashboards were developed, and advanced analytics modules covering VaR, CVaR, rolling Sharpe ratio, cohort analysis, investor risk segmentation, and a recommendation engine were implemented. The project delivers a robust analytical foundation that can be extended with real-time data and machine learning capabilities in future iterations.